

Perform Z-test for single population mean.

Problem statement:

A rice manufacturing company claims that the average weight of its rice bags is 50 kg. The population standard deviation of the bag weights is known to be 1.2 kg. An inspector randomly selects 30 rice bags and finds that their average weight is 49.2 kg.

Using Python and the `scipy.stats` library, perform a two-tailed one-sample Z-test at the 5% significance level ($\alpha = 0.05$) to determine whether there is sufficient evidence to reject the company's claim.

Your Python program should:

1. Display the null hypothesis (H_0) and alternative hypothesis (H_1).
2. Calculate and display the Z-statistic.
3. Calculate and display the p-value.
4. Compare the p-value with the significance level and display the appropriate decision:
Reject H_0 or Fail to Reject H_0 .
5. Display the final conclusion in the context of the problem.

The hypotheses to be tested are:

- $H_0: \mu = 50 \text{ kg}$
- $H_1: \mu \neq 50 \text{ kg}$

Expected output: The program should display the hypotheses, Z-statistic, p-value, decision, and

conclusion.